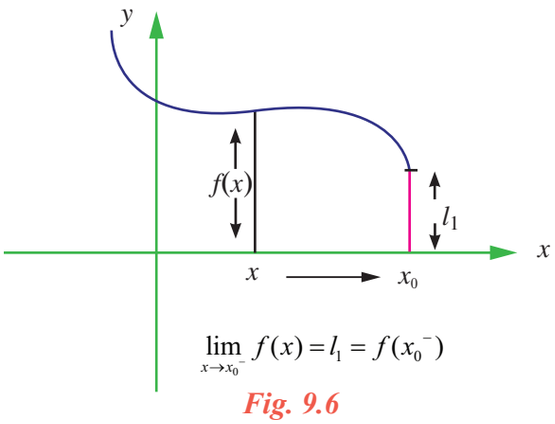
Definition 9.3

We say that the **right-hand limit** of $f(x)$ as x approaches x_0 (or the limit of $f(x)$ as x approaches x_0 from the right) is equal to l_2 if we can make the values of $f(x)$ arbitrarily close to l_2 by taking x to be sufficiently close to x_0 and greater than x_0. It is symbolically written as $f(x_0^+) = \lim_{x \to x_0^+} f(x) = l_2$.

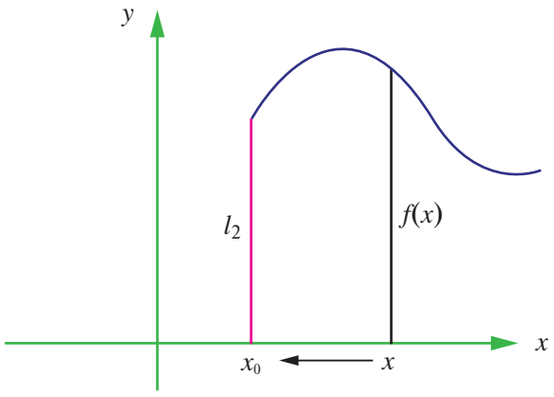
Thus the symbols "$x \to x_0^-$" and "$x \to x_0^+$" mean that we consider only $x < x_0$ and $x > x_0$ respectively.

These definitions are illustrated in the following Fig. 9.6 to 9.9.



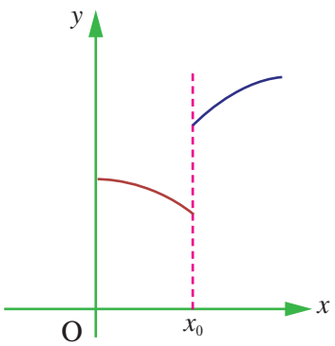
$\lim_{x \to x_0^-} f(x) = l_1 = f(x_0^-)$

Fig. 9.6



$\lim_{x \to x_0^+} f(x) = l_2 = f(x_0^+)$

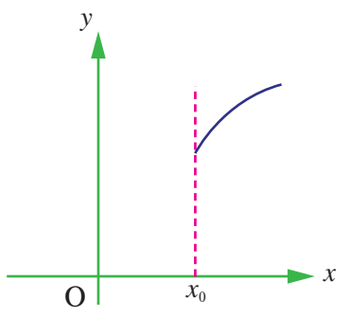
Fig. 9.7



$\lim_{x \to x_0} f(x)$ does not exist

Fig. 9.8

(Different values are obtained as x_0 is approached from the left and from the right)



$\lim_{x \to x_0} f(x)$ does not exist

Fig. 9.9

(Function not defined to the left of x_0)

From the above discussions we conclude that $\lim_{x \to x_0} f(x) = L$ exists if the following hold :

(i) $\lim_{x \to x_0^+} f(x)$ exists,

(ii) $\lim_{x \to x_0^-} f(x)$ exists and

(iii) $\lim_{x \to x_0^+} f(x) = \lim_{x \to x_0^-} f(x) = L$.

NLGZ5

From the definitions of one sided limits and that of the limit of $f(x)$ as we have the following :

$\lim_{x \to x_0} f(x) = L$ iff $\lim_{x \to x_0^-} f(x) = L = \lim_{x \to x_0^+} f(x)$.

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